

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Code of Conduct

I B Hemanth(192472127) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B Hemanth

Annexure A – Agile Sprint Planning & User Story Workshop

Project Title:

Telemedicine Platform for virtual doctor consultations

1. Project Vision and Stakeholder Identification

Project Vision

The vision of the **Telemedicine Platform for Virtual Doctor Consultations** is to provide a secure and user-friendly digital healthcare platform that enables patients to consult qualified doctors remotely through video calls, voice calls, and online chat. Many patients face difficulties in accessing healthcare services due to long travel distances, busy schedules, or emergencies. The proposed system addresses these challenges by connecting patients and healthcare professionals through a reliable online consultation platform.

The project aims to simplify healthcare delivery by providing appointment booking, electronic prescriptions, medical record management, secure online payments, and real-time communication. By reducing waiting time and improving accessibility, the platform enhances patient care while enabling doctors to provide quality healthcare services from anywhere. The system also promotes continuous healthcare monitoring and supports better medical decision-making through digital health records.

Project Objectives

The primary objectives of the project are:

- Enable patients to consult doctors remotely through video, voice, and chat.
- Allow patients to book appointments with available doctors..

- Provide secure user registration and authentication.
- Maintain digital medical records for patients.
- Generate electronic prescriptions after consultation.
- Support secure online payment for consultations.

Stakeholder	Role in the Project
Patient	Register, search doctors, book appointments, attend consultations, and view prescriptions..
Doctor	Conduct online consultation, manage appointments, prescribe medicines
Clinic Administrator	Manage doctors, monitor appointments, and supervise healthcare services.
System Administrator	Manage users, monitor application performance, maintain database, and ensure security.
Payment Gateway	Handle secure payment processing and transaction verification.
Notification Service	Send reminders, consultation alerts, and prescription notifications.

Stakeholders

Stakeholder Responsibilities

Patients

- Register and log in securely.
- Create and manage personal profiles.
- Search doctors by specialization.
- Book online appointments.

Doctors

- Register and verify professional credentials.
- Manage availability schedule.
- Accept or reject appointments.
- Conduct virtual consultations.

Clinic Administrator

- Register doctors and departments.
- Manage doctor schedules.
- Monitor appointments.
- Maintain hospital information. Recruiters.

Payment Gateway

- Process secure online payments.
- Verify payment transactions.

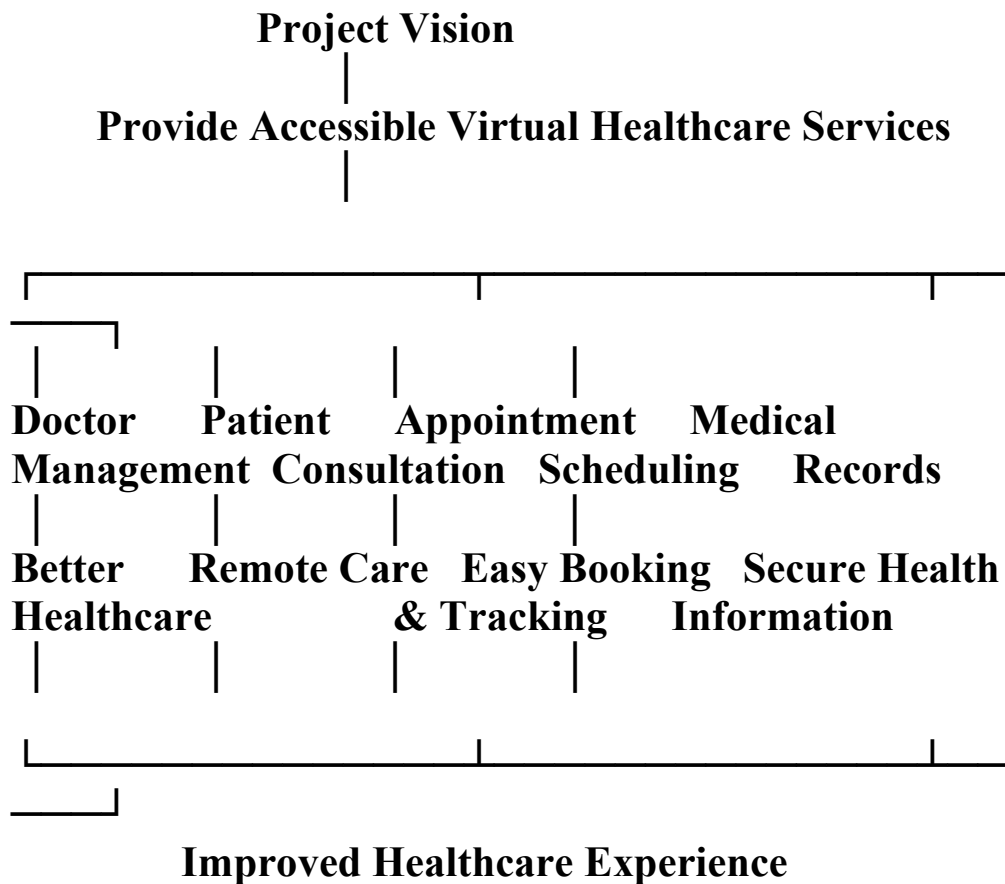
- Handle payment failures.
- Generate digital payment receipts.

System Administrator

- Manage users and permissions.
- Monitor application performance.
- Maintain databases.

Notification Service

- Send appointment confirmations.
- Send appointment reminders.



2. Epics & User Stories

Introduction

2.1 In Agile software development, an **Epic** represents a large feature or business functionality that can be divided into multiple user stories. **User Stories** describe system requirements from the end user's perspective using the format "**As a...**, **I want...**, **so that...**". They help the development team understand user needs, prioritize features, and deliver the project incrementally through sprints.

Project Epics

Epic ID	Epic Name	Description
EP-1	User Authentication	Allow patients and doctors to securely register and log in using email, mobile number, or OTP authentication.
EP-2	User Profile Management	Enable patients and doctors to create, update, and manage their personal and professional profiles.
EP-3	Doctor Search & Appointment Booking	Allow patients to search doctors by specialization, location, ratings, and book appointments online.
EP-4	Virtual Consultation	Enable secure video, voice, and chat consultations between doctors and patients.
EP-5	Electronic Medical Records	Store and manage patient medical history, prescriptions, reports, and consultation records securely.
EP-6	Dashboard & Notifications	Display appointment schedules, consultation history, reminders, and health notifications.
EP-7	Payment & Prescription Management	Support secure online payments and digital prescription generation after consultations.
EP-8	Administration & System Management	Manage users, doctors, appointments, reports, and maintain system security and performance.

2.2 User Stories

Story ID	Epic	User Story
US-01	EP-1	As a patient, I want to register and log in securely, so that I can access the telemedicine platform.
US-02	EP-2	As a patient, I want to update my personal and medical profile, so that doctors can understand my medical history..
US-03	EP-2	As a doctor, I want to manage my profile, specialization, and availability, so that patients can book appointments easily.
US-04	EP-3	As a patient, I want to search doctors based on specialization and ratings, so that I can find the most suitable healthcare professional.
US-05	EP-3	As a patient, I want to book, reschedule, or cancel appointments online, so that I can manage my consultations conveniently.
US-06	EP-3	As a patient, I want to view available consultation slots, so that I can choose a suitable appointment time.

Story ID	Epic	User Story
US-07	EP-4	As a patient, I want to attend virtual consultations through video or voice calls, so that I can receive medical advice remotely.
US-08	EP-5	As a doctor, I want to conduct online consultations, so that I can provide healthcare services from anywhere.
US-09	EP-5	As a doctor, I want to access patient medical records, so that I can make informed treatment decisions.
US-10	EP-6	As a patient, I want to receive appointment reminders and notifications, so that I never miss my consultation.
US-11	EP-6	As a patient, I want to view my consultation history and health records, so that I can monitor my healthcare progress.
US-12	EP-7	As a patient, I want to view my consultation history and health records, so that I can monitor my healthcare progress.
US-13	EP-7	As a patient, I want to pay consultation fees securely online, so that I can complete my appointment without visiting the hospital..
US-14	EP-8	As an administrator, I want to manage user accounts and doctor registrations, so that the system remains secure and organized.
US-15	EP-8	As an administrator, I want to monitor application performance and user activity, so that I can maintain system reliability.
US-16	EP-7	As a patient, I want to download my digital prescriptions and medical reports, so that I can use them for future treatments.

2.3 Mapping of Epics to User Stories

Epic	Associated User Stories
EP-1 – User Authentication	US-01
EP-2 – User Profile Management	US-02, US-03
EP-3 – Doctor Search & Appointment Booking	US-04, US-05, US-06
EP-4 – Virtual Consultation	US-07
EP-5 – Electronic Medical Records	US-08, US-09
EP-6 – Dashboard & Notifications	US-10, US-11
EP-7-Payment & Prescription Management	US-12, US-13, US-16
EP-8 – Administration & System Management	US-14, US-15

3. Acceptance Criteria				
Introduction				
Acceptance Criteria define the conditions that must be satisfied before a user story is considered complete. They help developers, testers, and stakeholders understand the expected behavior of each feature. In Agile Scrum, the Given–When–Then format is commonly used to create clear, measurable, and testable acceptance criteria.				
Acceptance Criteria for User Stories				
US-01: GitHub Login				
User Story				
As a beginner developer,				
I want to log in using my GitHub account,				
So that I can securely access the application.				
<table><tr><td>Acceptance Criteria (Given – When – Then)</td></tr><tr><td>Given the user has a valid GitHub account</td></tr><tr><td>When the user clicks the Login with GitHub button</td></tr><tr><td>Then the system should authenticate the user successfully and redirect them to the dashboard</td></tr></table>	Acceptance Criteria (Given – When – Then)	Given the user has a valid GitHub account	When the user clicks the Login with GitHub button	Then the system should authenticate the user successfully and redirect them to the dashboard
Acceptance Criteria (Given – When – Then)				
Given the user has a valid GitHub account				
When the user clicks the Login with GitHub button				
Then the system should authenticate the user successfully and redirect them to the dashboard				
US-02: Update User Profile				
User Story				
As a beginner developer,				
I want to update my programming skills and interests,				
So that I receive personalized recommendations.				
<table><tr><td>Acceptance Criteria (Given – When – Then)</td></tr></table>	Acceptance Criteria (Given – When – Then)			
Acceptance Criteria (Given – When – Then)				

Given the user is logged in
When the user updates their skills and clicks Save
Then the profile should be updated successfully and used for future recommendations

US-03: Edit Profile

Acceptance Criteria (Given – When – Then)
Given the user is viewing their profile
When the user edits profile information and saves it

Acceptance Criteria (Given – When – Then)	
Then the updated information should be stored successfully	

US-04: Repository Recommendation

Acceptance Criteria (Given – When – Then)	
Given the user has completed their profile	
When the system retrieves repository information from GitHub	
Then beginner-friendly repositories matching the user's skills should be displayed	

US-05: Repository Filter

Acceptance Criteria (Given – When – Then)	
Given repository recommendations are displayed	
When the user selects a programming language filter	
Then only repositories matching that language should be shown	

US-06: View Good First Issues

Acceptance Criteria (Given – When – Then)	
Given repositories are available	
When the user selects Good First Issue	
Then the system should display only beginner-friendly issues	

US-07: GitHub API Integration

Acceptance Criteria (Given – When – Then)	
Given GitHub API is available	
When repository data is requested	
Then the latest repositories, issues, and pull requests should be retrieved successfully	

US-08: Contribution Tracking

Acceptance Criteria (Given – When – Then)	
Given the user has contributed to a repository	
When GitHub updates the contribution	
Then the system should display the updated contribution history	

US-09: Pull Request Status

Acceptance Criteria (Given – When – Then)	
Given a pull request exists	
When the system synchronizes with GitHub	
Then the latest pull request status (Open, Closed, or Merged) should be displayed	
US-10: Dashboard	
Acceptance Criteria (Given – When – Then)	

Acceptance Criteria (Given – When – Then)	
Given contribution data is available	
When the dashboard is opened	
Then contribution statistics and learning progress should be displayed	
US-11: Analytics	
Acceptance Criteria (Given – When – Then)	
Given contribution records are stored	
When analytics are generated	
Then repository statistics and technology usage should be displayed correctly	
US-12: Portfolio Generation	
Acceptance Criteria (Given – When – Then)	
Given the user has contribution history	
When the user clicks Generate Portfolio	
Then the system should create a professional portfolio containing contribution details	

4. Product Backlog Prioritization

Introduction

A Product Backlog is an ordered list of all the features, enhancements, and functionalities required to develop the software system. Each user story is prioritized according to its business value, customer needs, and dependencies. High-priority items are implemented first so that the core functionality of the application becomes available as early as possible.

4.1 Product Backlog

Backlog ID	User Story	Priority	Business Value	Dependency
US-01	User registration and login	High	Users cannot access the application without authentication.	None
US-02	Create and update user profile	High	Required for personalized recommendations.	US-01
US-03	Edit profile information	High	Allows users to keep their information updated.	US-02
US-04	Recommend beginner-friendly repositories	High	Core functionality of the application.	US-02, US-07
US-05	Filter repositories by programming language	High	Improves repository search experience.	US-04
US-06	Display "Good First Issue" tasks	High	Helps beginners identify suitable issues.	US-07
US-07	Retrieve repository data using GitHub API	High	Required to fetch repository information.	US-01

Backlog ID	User Story	Priority	Business Value	Dependency
US-08	Track pull requests and issues	High	Enables contribution tracking.	US-07
US-09	Display pull request status	Medium	Provides contribution updates.	US-08
US-10	Contribution Dashboard	High	Displays user progress and analytics.	US-08
US-11	Technology usage analytics	Medium	Helps users analyze learning progress.	US-10
US-12	Generate contribution portfolio	Medium	Useful for placements and internships.	US-10
US-13	Notification system	Low	Additional convenience feature.	US-04
US-14	Manage users	Medium	Required for administration.	US-01
US-15	Monitor application performance	Low	Helps maintain system reliability.	US-14
US-16	Recruiter portfolio view	Low	Additional feature for recruiters.	US-12

4.2 Prioritization Strategy

The backlog is prioritized using three important factors:

Business Value

Features that provide the greatest benefit to users, such as authentication, repository recommendation, and contribution tracking, are given the highest priority.

Customer Needs

The application is designed primarily for beginner developers. Therefore, features that directly improve their experience are developed before secondary functionalities.

Dependencies

Some modules depend on others. For example, repository recommendations require the user profile and GitHub API integration to be completed first. Therefore, dependent features are scheduled only after their prerequisites are finished.

5. Sprint Planning

Sprint Duration

- **Sprint Length:** 2 Weeks
- **Sprint Number:** Sprint 1
- **Team Size:** 4 Members (*or Individual Developer if applicable*)

Sprint Goal

Develop the core functionalities of the Open-Source Contribution Tracker by implementing secure GitHub authentication, user profile management, GitHub API

integration, and repository recommendation to enable beginners to discover suitable open-source projects.

5.1 Sprint Backlog

Task ID	Development Task	Mapped User Story	Owner	Status
T-01	Develop user registration and login module	US-01	Developer	Planned
T-02	Implement patient and doctor profile management	US-02	Developer	Planned
T-03	Create doctor search functionality	US-03	Developer	Planned
T-04	Develop appointment booking and scheduling	US-07	Developer	Planned
T-05	Implement virtual consultation module	US-07	Developer	Planned
T-06	Add appointment reminders and notifications	US-04	Developer	Planned
T-07	Integrate secure online payment gateway	US-05	Developer	Planned
T-08	Display Good First Issues	US-06	Developer	Planned
T-09	Unit Testing	All	Tester	Planned
T-10	Perform testing and fix identified bugs	All	Team	Planned

6. Sprint Goal & Expected Deliverables

6.1 Sprint Goal

The primary goal of Sprint 1 is to develop the core functionalities of the Open-Source Contribution Tracker and Recommendation System for Beginner Developers. By the end of the sprint, users should be able to securely log in using their GitHub account, create and manage their profile, retrieve repository information through the GitHub API, and receive beginner-friendly repository recommendations. These features establish the foundation for future sprints, which will include contribution tracking, analytics, portfolio generation, and notifications.

6.2 Expected Deliverables

At the completion of Sprint 1, the following deliverables are expected:

7. User registration and secure login module.
8. Patient and doctor profile management feature.
9. Doctor search and appointment booking module.
10. Virtual consultation (video/voice/chat) functionality.
11. Electronic medical records management.
12. Digital prescription generation feature.
13. Secure online payment integration.
14. Appointment reminders and notification system.

14.1 Definition of Done (DoD)

A user story is considered **Done** only when all the following conditions are satisfied:

The feature is fully implemented according to the project requirements.

- The code has been reviewed and follows coding standards.
- Unit testing is completed successfully.
- The feature works correctly without critical bugs.
- Patient and doctor data are stored securely.
- The functionality is integrated with the existing system.

6.4 Sprint Review

At the end of the sprint, the development team will demonstrate the completed features of the Telemedicine Platform for Virtual Doctor Consultations to the Product Owner and stakeholders. Feedback will be collected from patients, doctors, and administrators to verify that the implemented functionalities meet user requirements. Any improvement suggestions or new requirements identified during the review will be added to the Product Backlog for future sprints.

6.5 Sprint Retrospective

At the end of the sprint, the Scrum Team will reflect on the work completed and discuss what went well, the challenges faced, and areas for improvement. The team will identify action items to enhance collaboration, development practices, and sprint efficiency. These improvements will be implemented in the next sprint to ensure continuous delivery of a high-quality telemedicine platform.

Conclusion

The Agile Sprint Planning process provides a structured approach to developing the **Telemedicine Platform for Virtual Doctor Consultations**. By defining a clear project vision, identifying stakeholders, creating epics and user stories, establishing acceptance criteria, prioritizing the product backlog, and planning sprint goals, the project is organized into manageable and achievable iterations.

Using the Scrum framework enables the development team to deliver the platform incrementally while incorporating continuous feedback from patients, doctors, and administrators. This iterative approach improves collaboration, adaptability, and software quality, ensuring that the final system provides secure virtual consultations, efficient appointment management, digital prescriptions, electronic medical records, and reliable healthcare services. Ultimately, the platform enhances healthcare accessibility, reduces waiting time, and delivers a convenient, secure, and user-friendly telemedicine experience for all stakeholders.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25